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CPLN 6710: Statistical and Data Mining Methods for Urban Data Analysis

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Using Spatial Lag, Spatial Error and Geographically Weighted Regression to Predict Median House Values in Philadelphia Block Groups

1 – Introduction

Median house values are not distributed randomly across urban space. In Philadelphia, housing markets are shaped by neighborhood socioeconomic conditions, housing characteristics, and the influence of nearby areas, which means that block groups located near one another often share similar housing outcomes. In the previous report, an ordinary least squares (OLS) regression was used to examine variation in logged median house value (LNMEDHVAL) across Philadelphia Census block groups. The model used four predictors: logged number of households below poverty (LNNBELPOV), percent of residents with at least a bachelor's degree (PCTBACHMOR), percent of detached single-family housing units (PCTSINGLES), and percent of vacant housing units (PCTVACANT). This earlier analysis established that these neighborhood characteristics were significantly associated with median house values in Philadelphia.

However, OLS regression is often not fully appropriate for spatial data because one of its key assumptions is that observations and errors are independent. In a geographic setting such as Philadelphia block groups, this assumption is frequently violated because nearby places tend to influence one another and share spatially clustered unobserved characteristics. As a result, residual spatial autocorrelation may remain even after fitting an OLS model. The purpose of this report is therefore to extend the earlier analysis by applying three spatial methods: spatial lag regression, spatial error regression, and geographically weighted regression (GWR) to determine whether these approaches account for spatial dependence more effectively and provide a better fit than the original OLS model.

2 – Methods

2.1 – A Description of the Concept of Spatial Autocorrelation

Before creating and running our three spatial models, we mapped and analyzed the spatial autocorrelation of our dependent variable (LNMEDHVAL, the log of median home values). The concept of spatial autocorrelation derives from Tobler's First Law of Geography: "Everything is related to everything else, but near things are more related than distant things" (Tobler, 1970). To measure spatial autocorrelation, we use Moran's I which ranges from 1 (perfect clustering of similar values) to -1 (perfect dispersion, such as with a checkerboard). The formula is shown

below where I is the value of Moran's I , n is the number of observations, X is the variable being tested and $\bar{X}$ is its mean, X_i is the variable value at a particular location i , X_j is the variable value at another location j , and w_{ij} is a weight indexing location of i relative to j .

$$I = \frac{\left(\frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (X_i - \bar{X})(X_j - \bar{X})}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \right)}{\left(\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n} \right)} =$$

$$= \frac{n}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}} \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (X_i - \bar{X})(X_j - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Throughout this report, we used Queen-1 weights. This means that, for any given census block groups, we will only be considering the other blocks touching it (even at just corners) as neighbors. Since we're only using a contiguity of 1, we will not consider neighbors of neighbors. Some statisticians opt to compare spatial autocorrelation across different weights matrices to ensure that autocorrelation is actually occurring and not just an artifact of their weights definition, but we opted to skip this process given the main goal of this report is to compare spatial regression methods.

To test the significance of Moran's I , we must test whether or not we can reject the null hypothesis (H_0) that there is no spatial autocorrelation (i.e. variables are randomly distributed). The alternative hypotheses are H_{A1} , positive spatial autocorrelation, and H_{A2} , negative spatial autocorrelation. To do this, we first compute Moran's I for our variable (LNMEDHVAL). Next, we randomly shuffle the values of this variable 999 times and compute Moran's I again on the shuffled maps. Once we have these 1,000 observations of Moran's I , we list them in descending order (or ascending if it's negative) and compare where the original Moran's I value stands in the rankings in comparison to the Moran's I values for the random permutations. We use its place in this ranking to calculate a Pseudo P-value by taking the ranking of the original Moran's I and dividing it by the total number of observations (in this case, 1,000 as there is 1 original observation and 999 permutations). For example, if the Moran's I value for LNMEDHVAL is 1st out of 1,000, the Pseudo P-value would be 0.001, meaning that there is at most a 0.1% chance of observing a Moran's I at least as high as the original observation if there is no spatial autocorrelation present. If the Pseudo P-value is less than 0.05, we consider the spatial autocorrelation significant and reject H_0 of no spatial autocorrelation.

Thus far, this description of spatial autocorrelation applies to the global distribution of the variable, not how the variable behaves locally. Thinking back to Tobler's First Law of Geography, we want to also measure to what extent near things are (or are not) related. To do this, we use Local Indices of Spatial Autocorrelation (LISA) to map four types of clusters: (1) High-High, where areas with high values of LNMEDHVAL have neighbors that also have high values, (2) Low-Low, where areas with low values of LNMEDHVAL have neighbors that also have low values, (3) High-Low, where areas with high values of LNMEDHVAL have neighbors with low values, and (4) Low-High, where areas with low values of LNMEDHVAL have neighbors with high values. To test for significance in the Local LISA statistic (local Moran's I), we test the null hypothesis H_0 that there is no local spatial autocorrelation at location i . For Local LISA statistics, the majority of the process of significance calculating is the same as global Moran's I. The main difference is that we use permutations of the values of variable X among the locations in the dataset, with the value at location i being held constant (not permuted/shuffled). The value of I_i (Moran's I at location i) for the original dataset is ranked relative to the permuted values. When values of I_i for the original dataset are very low or very high compared to the permuted list, they are "of interest." We can obtain a Pseudo P-value by taking the rank of the actual value of I_i and dividing it by the total number of observations (including it) after permutations just like with Moran's I above.

2.2 – Review of OLS Regression and its Assumptions

Ordinary Least Squares (OLS) Regression is a statistical method used to examine the relationship between a given dependent variable (in this case, the log of median home values) and one (simple regression) or more (multiple regression) predictor variables. With OLS Regression, we can calculate the amount by which the dependent variable changes when a predictor variable increases by one unit, holding all other predictors constant. OLS Regression uses the least squares estimation to fit a surface in $k+1$ dimensions (where k is the number of predictors) by minimizing the sum of squared residuals between the points and the estimated surface. There are several assumptions that accompany Simple Regression, namely (1) there is a linear relationship between predictors and the dependent variable, (2) residuals are normally distributed, homoscedastic, random, and independent of observations, and (3) the dependent variable is continuous. For Multiple Regression, assumptions are the same and, as an addition, predictors should not be multicollinear (strongly correlated with each other). For more information on OLS regression, see our previous paper *Assignment 1: Exploring Median Home Value in Philadelphia Through OLS Regression*.

When working with data that has a spatial component (as we are in this report), the assumption that errors are random and independent often does not hold. We test this assumption on our LNMEDHVAL variable by examining spatial autocorrelation of the residuals using Moran's I. An alternative method for testing OLS residuals for spatial autocorrelation is to regress them on nearby residuals (in this analysis, these are the residuals that directly border the neighboring

block group in accordance with our Queen-1 weights matrix). Another assumption of OLS Regression that can be tested in GeoDa is homoscedasticity. To test the null hypothesis that residuals are homoskedastic (have constant variance), we enable the White test for heteroscedasticity in the OLS regression output in GeoDa. If the White test p-value is below 0.05, we reject the null hypothesis of homoskedasticity. For the assumption of normality of errors, OLS regression outputs in GeoDa include the Jarque-Bera test on normality of errors. The null hypothesis of the Jarque-Bera test is that errors are normally distributed and if the Jarque-Bera statistic is significantly larger than the degrees of freedom and has a P-value below 0.05, we can reject that null hypothesis.

2.3 – Spatial Lag and Spatial Error Regression

Spatial lag and spatial error regression models were estimated in R to account for spatial dependence that remained after the OLS analysis. Geographically structured datasets often share unmeasured spatial characteristics and influence one another, which violates the OLS assumption that errors are independent. Therefore, spatial regression models are applied to address this issue by considering spatial structure.

The spatial lag model assumes that the dependent variable in one location is directly influenced by values of the dependent variable in neighboring locations. In this analysis, the spatial lag model can be written as:

$$LNMEDHVAL_i = \rho \sum_j w_{ij} LNMEDHVAL_j + \beta_0 + \beta_1 LNNBELPOV_i + \beta_2 PCTBACHMOR_i + \beta_3 PCTSINGLES_i + \beta_4 PCTVACANT_i + \varepsilon_i$$

Where $LNMEDHVAL_i$ is the logged median house value in block group i , and

$\sum_j \omega_{ij} LNMEDHVAL_j$ is the spatially lagged dependent variable, representing the weighted average of neighboring block groups' logged median house values based on the Queen contiguity weights matrix. In this weight matrix, neighboring block groups are defined as those that share either a boundary or a corner. The coefficient ρ is the spatial autoregressive parameter and measures the strength of the relationship between a block group's house value and the house values of its neighbors. The term β_0 is the intercept. β_1 is the coefficient for LNNBELPOV, β_2 is the coefficient for PCTBACHMORE, β_3 is the coefficient for PCTSINGLES, and β_4 is the coefficient for PCTVACANT. Finally, ε_i is the random error term. A positive and statistically significant ρ would indicate that nearby block groups tend to have similar median house values even after accounting for the observed predictors.

The spatial error regression model was used to address a different type of spatial dependence. Rather than assuming that the dependent variable itself is directly influenced by neighboring values, it assumes that the unobserved influences captured in the error term are spatially correlated. In this analysis, the spatial error model can be written as:

$$LNMEDHVAL_i = \beta_0 + \beta_1 LNNBELPOV_i + \beta_2 PCTBACHMOR_i + \beta_3 PCTSINGLES_i + \beta_4 PCTVACANT_i + u_i$$

$$u_i = \lambda \sum_j w_{ij} u_j + \varepsilon_i$$

where $LNMEDHVAL_i$ is the logged median house value in block group i , β_0 is the intercept. β_1 is the coefficient for $LNNBELPOV$, β_2 is the coefficient for $PCTBACHMORE$, β_3 is the coefficient for $PCTSINGLES$, and β_4 is the coefficient for $PCTVACANT$. The term u_i is the spatially structured error component, which captures spatial dependence in the unobserved part of the model. The coefficient λ is the spatial error parameter and measures the strength of spatial autocorrelation in the error term. The term $\sum_j \omega_{ij} u_j$ represents the weighted average of neighboring error terms based on the Queen contiguity weights matrix, and ε_i is the remaining random error. A significant value of λ indicates that omitted influences on median house values are spatially clustered and that the spatial error model accounts for this dependence directly.

Both spatial lag and spatial error models require the same assumptions for OLS regression, except for the assumption of spatial independence. The difference is that spatial lag and spatial error regression explicitly allow geographic dependence that would otherwise violate the independence assumption of OLS.

The goal of using spatial lag and spatial error regression is to improve model specification when spatial dependence is present. Ideally, after incorporating spatial structure, the residuals should exhibit less spatial autocorrelation than the OLS residuals. In practical terms, these models are intended to reduce the clustering pattern that remains in the errors of a non-spatial regression model and to produce a better overall fit to the data.

To evaluate whether the spatial lag and spatial error models performed better than OLS, each spatial model was compared with the OLS model using several criteria. These included the Akaike Information Criterion (AIC), Schwarz Criterion (BIC), the log likelihood, and the likelihood ratio test. The AIC measures model fit against model complexity; lower values indicate a better model. The log likelihood measures how well the model fits the observed data, with higher values indicating a better fit. The likelihood ratio test compares a spatial model to its corresponding OLS model and evaluates whether the improvement in fit is statistically significant. In that test, the null hypothesis is that the spatial model does not improve fit beyond OLS, while the alternative hypothesis is that the spatial model provides a significantly better fit. A statistically significant likelihood ratio test therefore, supports the spatial specification over OLS.

Another important way to compare the OLS, spatial lag, and spatial error models is to examine Moran's I of the regression residuals. If a spatial model is working well, the Moran's I value for its residuals should be closer to zero than that of the OLS residuals, and ideally it should no

longer be statistically significant. Thus, when comparing models using residual Moran's I, the preferred model is the one that leaves the least residual spatial autocorrelation while also maintaining a strong overall model fit. Together, these criteria make it possible to assess whether the spatial regression models provide a meaningful improvement over the original OLS specification.

2.4 – Geographically Weighted Regression

The Geographically Weighted Regression (GWR) will be run in R, along with the map creation. GWR is a method of spatial regression that seeks to address the issue described in the Simpson's paradox in which one can fit a regression on a dataset and find the overall trend, but it may totally miss trends unique to different groups within the dataset. For example one could regress the correlation between burglaries and median house values across all of Philadelphia, but would not see that burglaries affect median house value in South Philadelphia differently than it does in Northeast Philadelphia.

The GWR equation is similar to that of OLS, but is repeated for each location in the regression, and the relationship between the independent and dependent variable is specific to that location. The equation is written as follows:

$$y_i = \beta_{i0} + \beta_{i1}x_{i1} + \beta_{i2}x_{i2} + \dots + \beta_{im}x_{im} + \varepsilon_i = \beta_{i0} + \sum_{k=1}^m \beta_{ik}x_{ik} + \varepsilon_i$$

Where i represents the specific observation (location), m represents the total number of predictor variables in the model, and $k = 1$ through m . By finding coefficients for each individual observation, GWR is primarily used to model spatial nonstationarity, though it may also reduce residual spatial autocorrelation. To run GWR, there must be multiple locations. GWR gives greater weights to observations closer to observation i , following Tobler's Law which states that "near things are more related than distant things". To weigh nearby locations, you may use fixed bandwidth in which the number of observations may vary but the bandwidth area remains constant, or an adaptive bandwidth in which the number of observations stays consistent but the bandwidth area changes. In this regression, we will be using an adaptive bandwidth as it is the more appropriate type when point distribution varies across space. Philadelphia Census Block Groups are not spatially uniform, thus our point distribution varies across space.

Many of the same assumptions relevant to OLS remain important in GWR, including concerns about residual distribution, heteroskedasticity, and multicollinearity. In GWR, residuals should be normal, there should be homoscedasticity, and there should be no multicollinearity. GWR requires at least 300 observations to run, which our dataset meets with 1720 observations. In GWR, multicollinearity can be an issue when multiple predictors have clusters of high or low values in the same area. In the results of a GWR, the condition number indicates whether results

are unstable due to multicollinearity. Generally, features with a condition number greater than 30 are not trustworthy.

In a GWR output, p-values are not included since it would require multiple significance tests per observation, and by testing each of our 1720 observations at $p < 0.05$, we would likely see many false positives by chance alone. GWR's increased complexity would require thousands of tests.

3 – Results

3.1 – Spatial Autocorrelation Results

In running global Moran's I on LNMEDHVAL, we find a Moran's I value of 0.794 (suggesting that the log of median home values is strongly spatially autocorrelated). Figure 1 shows the Moran's I scatterplot of LNMEDHVAL using Queen-1 contiguity weights. Further, figure 2 shows the significance test of this value across 999 permutations, which produced a pseudo p-value of 0.001 (meaning there is a 0.1% chance of observing a global Moran's I value at least as high as 0.794 if no spatial autocorrelation is present). With both of these metrics combined, we can conclude that LNMEDHVAL is significantly spatially autocorrelated.

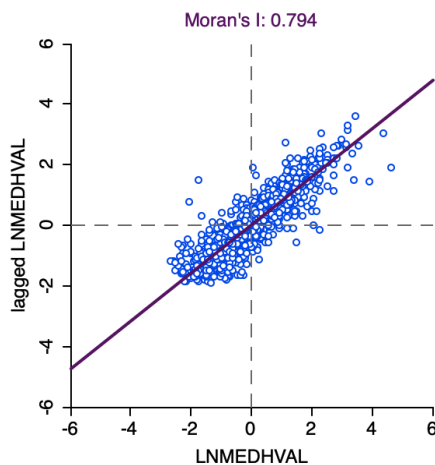


Figure 1: Global Moran's I and Scatter Plot for LNMEDHVAL (Queen 1 Contiguity Weight).

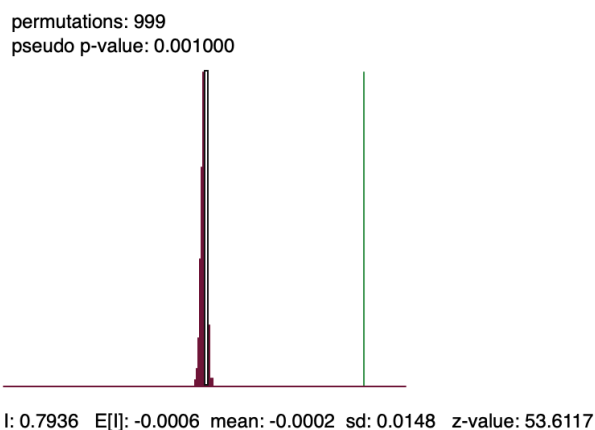


Figure 2: Global Moran's I Significance Test With Original Dataset Moran's I Value, Histogram of Moran's I Values Across 999 Permutations, and Pseudo P-Value for LNMEDHVAL (Queen 1 Contiguity Weight).

After determining that there is spatial autocorrelation occurring in LNMEDHVAL, we used Local Moran's I to see where the significant clusters of spatial autocorrelation are. Figure 3A displays a map of these results. We find that LNMEDHVAL has several "hotspots" (areas where the logged median home value is high, surrounded by neighbors with high logged median home values) across Philadelphia. These are northeast Philadelphia outside of the Northeast Corridor (areas surrounding Northeast Philadelphia Airport), Northwest Philadelphia (Roxborough, Manayunk, and Mt. Airy), Center City, and University City. Further, there are several "coldspots" where low logged median home values are surrounded by neighbors with low logged median home values. These are concentrated around North Philadelphia, Mantua, Grays Ferry, and Kingsessing. There are quite a few areas where the observed local Moran's I was not extreme enough relative to the permutation distribution to reject the local null (the pseudo p-value is greater than 0.05, meaning the actual value of local Moran's I for that location was not ranked 5 or higher when placed among 999 random permutations of the data). When you compare Figure 3A to a racial map of the city (Figure 3B), the trend is clear: hotspots of logged median home values are almost exclusively located in areas of white plurality whereas coldspots are almost exclusively located within BIPOC plurality areas.

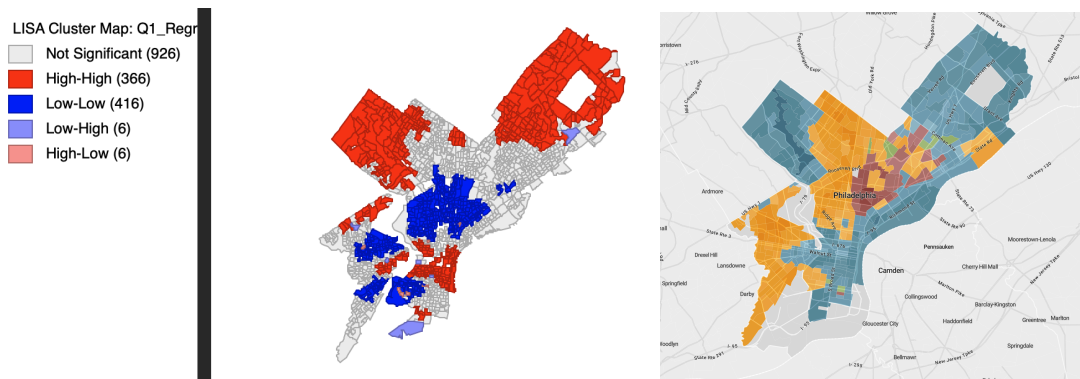


Figure 3A: Map of Local Spatial Autocorrelation Clusters for Values of LNMEDHVAL in Philadelphia Block Groups. Figure 3B: 2024 Racial Distribution Map of Philadelphia Tracts (Blue is white plurality, Green is Asian plurality, orange is Black plurality, and maroon is Hispanic plurality, darker colors indicate a larger plurality or majority).

3.2 – A Review of OLS Regression and Assumptions: Results

Table 1 shows the OLS Regression output from GeoDa where the dependent variable is LNMEDHVAL and the predictors are (1) the percentage of people over 25 years old with a bachelor's degree, (2) percentage of single-family detached housing units, (3) percentage of vacant housing units, and (4) the logged count of people living below the poverty line. All predictors here appear to be significant (having p-values below 0.05). The model has an

R-squared of 0.6623, meaning that 66.23% of the variance in logged median home value is explained by the model.

```

REGRESSION
-----
SUMMARY OF OUTPUT: ORDINARY LEAST SQUARES ESTIMATION
Data set      : RegressionData
Dependent Variable : LNMEDHVAL  Number of Observations: 1720
Mean dependent var : 10.882    Number of Variables : 5
S.D. dependent var : 0.62972   Degrees of Freedom : 1715
R-squared      : 0.662300     F-statistic : 840.869
Adjusted R-squared : 0.661513 Prob(F-statistic) : 0
Sum squared residual: 230.332 Log likelihood : -711.493
Sigma-square    : 0.134304 Akaike info criterion : 1432.99
S.E. of regression : 0.366475 Schwarz criterion : 1460.24
Sigma-square ML : 0.133914
S.E of regression ML: 0.365942
-----
Variable      Coefficient      Std.Error      t-Statistic      Probability
-----
CONSTANT      11.1138      0.0465318      238.843      0.00000
LNNBELPOV     -0.0789035   0.0084567      -9.3303      0.00000
PCTBACHMOR    0.0209095   0.000543184     38.4944      0.00000
PCTSINGLES    0.00297695   0.000703155      4.23371      0.00002
PCTVACANT     -0.0191563   0.000977851     -19.5902      0.00000
-----
REGRESSION DIAGNOSTICS
MULTICOLLINEARITY CONDITION NUMBER 12.990609
TEST ON NORMALITY OF ERRORS
TEST      DF      VALUE      PROB
Jarque-Bera 2      778.9646     0.00000
DIAGNOSTICS FOR HETEROSKEDASTICITY
RANDOM COEFFICIENTS
TEST      DF      VALUE      PROB
Breusch-Pagan test 4      162.9108     0.00000
Koenker-Bassett test 4      61.6992      0.00000
SPECIFICATION ROBUST TEST
TEST      DF      VALUE      PROB
White     14      111.3224     0.00000
DIAGNOSTICS FOR SPATIAL DEPENDENCE
FOR WEIGHT MATRIX : Q1_RegressionData
(row-standardized weights)
TEST      MI/DF      VALUE      PROB
Moran's I (error) 0.3129     22.3664     0.00000
Lagrange Multiplier (lag) 1      930.1626     0.00000
Robust LM (lag) 1      441.1061     0.00000
Lagrange Multiplier (error) 1      490.5691     0.00000
Robust LM (error) 1      1.5126       0.21875
Lagrange Multiplier (SARMA) 2      931.6751     0.00000

```

Table 1: OLS Regression Output (LNMEDHVAL, Queen 1 Contiguity Weight).

The output in Table 1 also shows the results of several tests for heteroskedasticity in this model: the Breusch-Pagan test, Koenker-Bassett test, and White test. All three tests show significance, with p-values of 0.00000, meaning that all three of them agree that we can reject the null hypothesis of homoskedasticity. When keeping this regression output and the results of the three tests for heteroskedasticity in mind, the scatterplot of standardized residuals and predicted values

(Figure 4) further confirms the presence of heteroskedasticity. The datapoints do not appear randomly distributed around zero, but instead appear slightly quadratic.

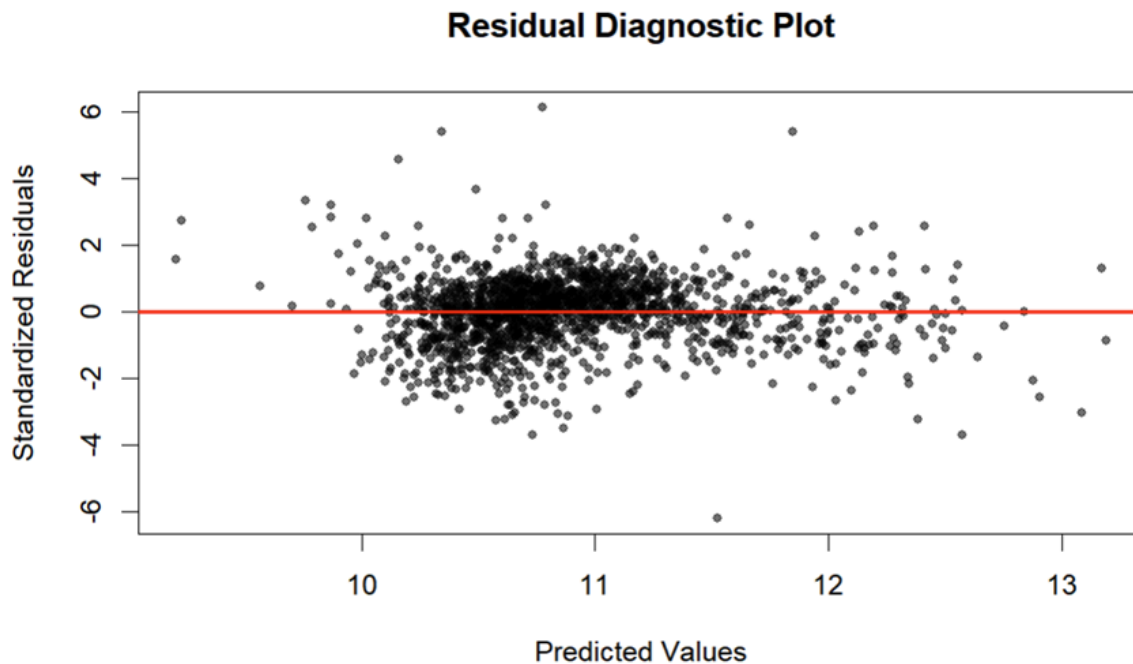


Figure 4: Plot of Standardized Residuals Compared to Predicted Values (From Homework 1).

Another test result seen in the output from Table 1 is the Jarque-Bera test for normality of errors. The result for this test is very significant, with a p-value of 0.00000, so we can reject the null hypothesis of this test in favor of the alternative hypothesis that errors are not normally distributed. This result is notably in opposition to our conclusions about the histogram of standardized residuals (Figure 5) that we analyzed in Homework 1, which generally appears to somewhat resemble a normal distribution. Given that the Jarque-Bera test is a more robust way of determining normality of errors than the human eye, this test could very well be picking up on the leaning of this model to produce more negative errors (below zero) than positive ones, with a slight bump in error frequency between 2.5 and 3.

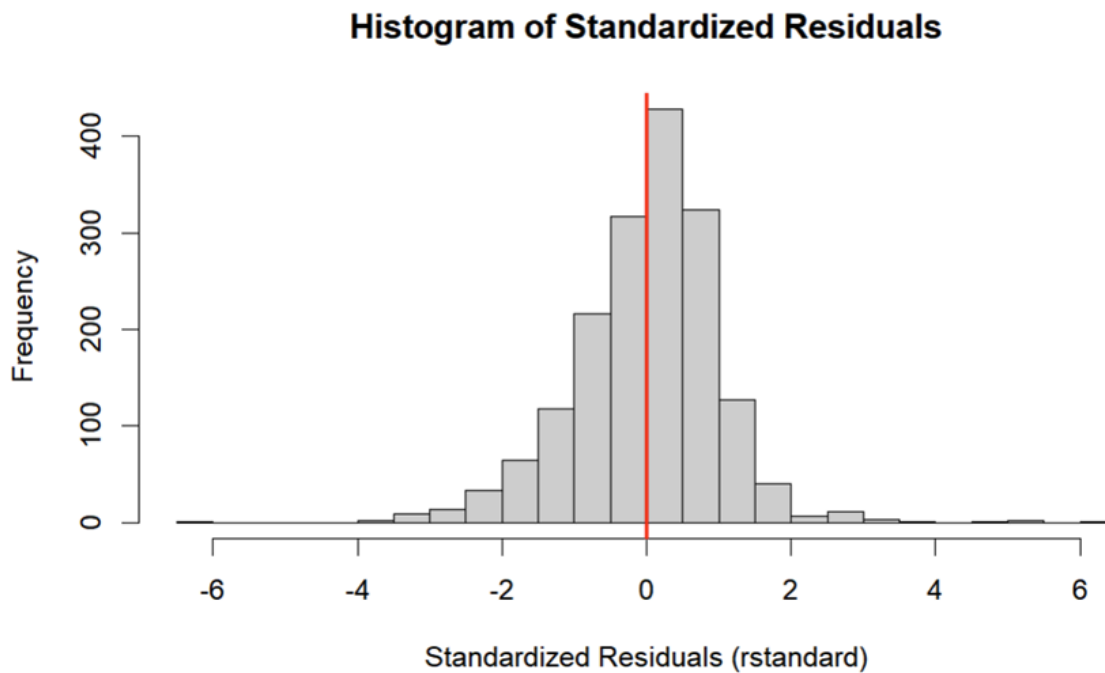


Figure 5: Histogram of standardized residuals for our model.

Figure 6 shows a scatter plot of OLS residuals compared to spatially lagged residuals using Queen-1 contiguity weights. One key metric for determining the presence of spatial autocorrelation in our residuals is to plot them against the spatially lagged residuals as we have done below. A key metric displayed in the statistics of this output is “slope b,” which is equivalent to the Beta coefficient of the spatially lagged residuals when regressing OLS residuals on the spatially lagged residuals. Here, we see a value of 0.733, which means that a one-unit increase in spatially lagged residuals is associated with a 0.733-unit increase in OLS residuals. This slope is statistically significant, with a p-value reported as < 0.001 , indicating spatial autocorrelation in our residuals.

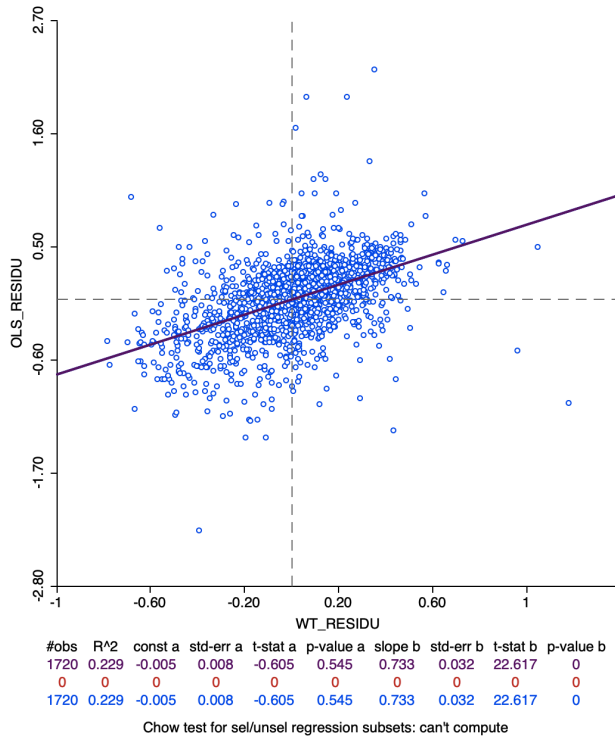


Figure 6: Scatter Plot of OLS Residuals (Y Axis, OLS_RESIDU) and Spatially Lagged Residuals (X Axis, WT_RESIDU).

The spatial autocorrelation of OLS residuals is further confirmed by running global Moran's I on the residuals. Both this metric and slope- b end up telling the same story of spatial autocorrelation of the residuals. The global Moran's I result, shown in Figure 7, has a value of $I = 0.313$. We test this result on 999 permutations (see Figure 8) and get a pseudo p-value of 0.001. This pseudo p-value tells us that I of 0.313 is statistically significant (as it is less than 0.05) and we therefore reject the null hypothesis that the distribution of residuals are completely random in favor of H_{A1} that there is positive spatial autocorrelation of residuals.

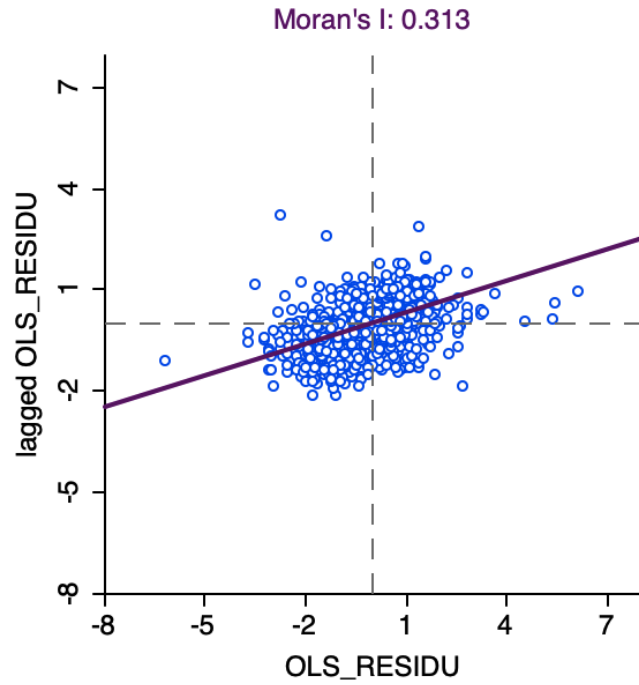
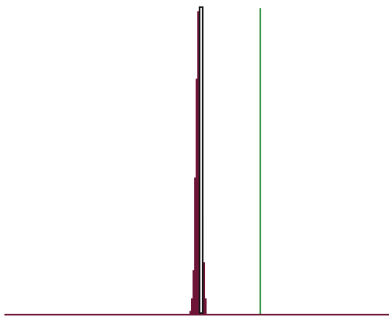


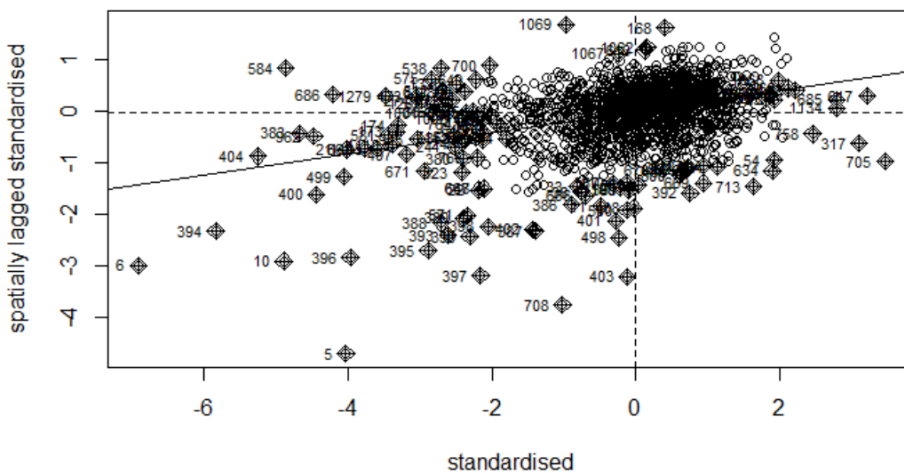
Figure 7: Global Moran's I and Scatter Plot for OLS Residuals (Queen 1 Contiguity Weight).

permutations: 999
pseudo p-value: 0.001000



I: 0.3129 E[I]: -0.0006 mean: -0.0007 sd: 0.0142 z-value: 22.0680

Figure 8: Global Moran's I Significance Test With Original Dataset Moran's I Value, Histogram of Moran's I Values Across 999 Permutations, and Pseudo P-Value for OLS Residuals (Queen 1 Contiguity Weight).



OLS Moran's I scatter plot for comparison below (generated in R)

3.3 – Spatial Lag and Spatial Error Regression Results

Spatial Lag Regression Results

Table 2 - Spatial Lag

```
Call:lagsarlm(formula = LNMEDHVAL ~ LNNBELPOV + PCTBACHMOR + PCTSINGLES +
  PCTVACANT, data = shp, listw = queenlist)
```

Residuals:				
Min	1Q	Median	3Q	Max
-1.655421	-0.117248	0.018654	0.133126	1.726436

Type: lag

Coefficients: (asymptotic standard errors)

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.89845489	0.20111357	19.3843	< 0.00000000000000022
LNNBELPOV	-0.03405466	0.00629287	-5.4116	0.00000006246
PCTBACHMOR	0.00851381	0.00052193	16.3120	< 0.00000000000000022
PCTSINGLES	0.00203342	0.00051577	3.9425	0.0008063503
PCTVACANT	-0.00852940	0.00074367	-11.4694	< 0.00000000000000022

Rho: 0.6511, LR test value: 911.51, p-value: < 0.000000000000000222

Asymptotic standard error: 0.01805

```
z-value: 36.072, p-value: < 0.000000000000000222
```

Wald statistic: 1301.2, p-value: < 0.000000000000000222

Log likelihood: -255.74 for lag model

ML residual variance (sigma squared): 0.071948, (sigma: 0.26823)

Number of observations: 1720

Number of parameters estimated: 7

AIC: 525.48, (AIC for lm: 1435)

LM test for residual autocorrelation

test value: 67.737, p-value: 0.00000000000000022204

Likelihood ratio for spatial linear models

```

data:
Likelihood ratio = 911.51, df = 1, p-value < 0.00000000000000022
sample estimates:
Log likelihood of lagreg      Log likelihood of reg
      -255.7400                -711.4933

[1] 525.48
[1] 563.6305
'log Lik.' -255.74 (df=7)

Breusch-Pagan test

data:
BP = 210.76, df = 4, p-value < 0.00000000000000022

studentized Breusch-Pagan test

data:
BP = 51.411, df = 4, p-value = 0.0000000001832

Jarque Bera Test

data: lagreg$residuals
X-squared = 2756.9, df = 2, p-value < 0.00000000000000022

```

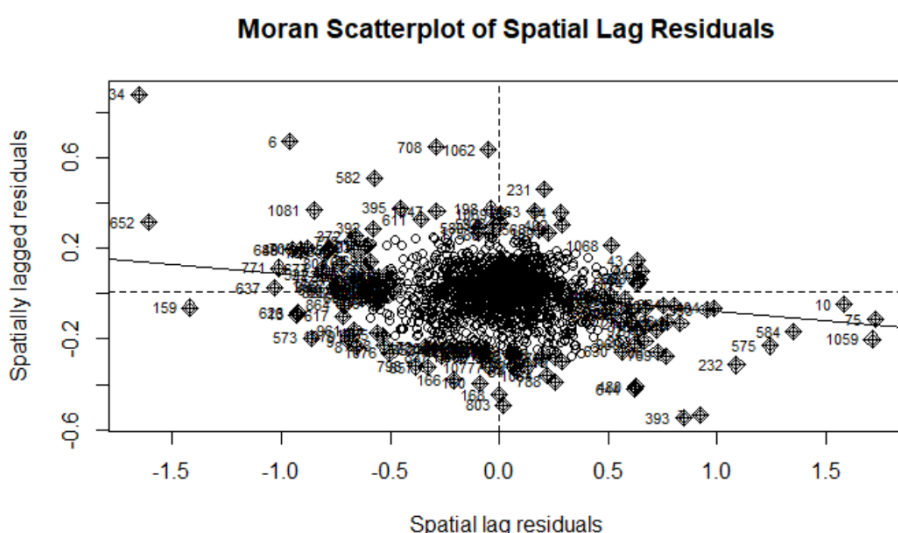
Table 2 presents the spatial lag regression results. In the R output, the spatially lagged dependent variable term is represented by the spatial autoregressive coefficient ρ , rather than appearing as a separate coefficient labeled `W_LNMEDHVAL`. In this model, $\rho = 0.6511$ and is highly statistically significant ($p < 0.001$), indicating strong positive spatial dependence in median house values across neighboring block groups. This means that median house values in one block group are associated with the weighted average median house values of nearby block groups, even after controlling for poverty, educational attainment, housing type, and vacancy. The significance of ρ indicates that spatial dependence is an important part of the housing value process and should be modeled explicitly rather than left in the residuals.

All four predictors remained statistically significant in the spatial lag model. `LNNBELPOV` had a negative coefficient ($\beta = -0.0341$, $p < 0.001$), indicating that higher levels of poverty were associated with lower median house values. `PCTBACHMOR` had a positive coefficient ($\beta = 0.0085$, $p < 0.001$), suggesting that block groups with higher proportions of residents holding at least a bachelor's degree tended to have higher median house values. `PCTSINGLES` was also positive and significant ($\beta = 0.0020$, $p < 0.001$), indicating that a higher share of detached single-family housing was associated with higher house values. `PCTVACANT` had a negative coefficient ($\beta = -0.0085$, $p < 0.001$), indicating that higher vacancy rates were associated with lower median house values. Compared with the OLS model, the directions of these relationships remained consistent, but the spatial lag model provides estimates after explicitly accounting for spatial dependence in the dependent variable.

Compared with the OLS model, the directions of these relationships remained unchanged, and all four predictors were significant in both models. However, the coefficients in the spatial lag model were smaller in absolute magnitude than in OLS. In the OLS model, the coefficients were -0.0789 for LNNBELPOV, 0.0209 for PCTBACHMOR, 0.0030 for PCTSINGLES, and -0.0192 for PCTVACANT, whereas in the spatial lag model, they were -0.0341, 0.0085, 0.0020, and -0.0085, respectively. This reduction in magnitude suggests that part of the association estimated by OLS reflected spatial dependence in housing values, which is captured more directly through the spatial lag term in the spatial lag specification.

The Breusch-Pagan test indicates that heteroskedasticity remained present in the spatial lag residuals. The standard Breusch-Pagan test was highly significant (BP = 210.76, $p < 0.001$), and the studentized Breusch-Pagan test was also significant (BP = 51.411, $p < 0.001$). Thus, although the spatial lag model improves overall fit, it does not fully resolve heteroskedasticity. The Jarque-Bera test was likewise highly significant ($X^2 = 2756.9$, $p < 0.001$), indicating that the residuals were still not normally distributed. These results indicate that although the spatial lag model improves on OLS, it does not fully resolve all remaining diagnostic issues.

Compared with the OLS model, the spatial lag model showed a much better fit. Its AIC decreased from 1432.99 in the OLS model to 525.48 in the spatial lag model, indicating a substantial improvement. The Schwarz Criterion (BIC) also decreased from 1460.24 in OLS to 563.63 in the spatial lag model, again favoring the spatial lag specification. The log likelihood improved sharply from -711.493 in the OLS model to -255.740 in the spatial lag model. In addition, the likelihood ratio test comparing the spatial lag model to OLS was highly significant (LR = 911.5, $p < 0.001$), confirming that the spatial lag model provides a statistically meaningful improvement over the non-spatial OLS specification.



The Moran scatterplot of the spatial lag residuals indicates substantially less spatial autocorrelation than was present in the OLS residuals. Compared with the OLS plot, the residuals from the spatial lag model are much more tightly clustered around zero and the fitted relationship is much flatter, suggesting that the spatial lag specification absorbed a large share of the positive spatial dependence that remained in the OLS errors. This visual pattern is consistent with the Monte Carlo Moran's I test for the spatial lag residuals, which produced a value of -0.0824 with a p-value of 0.002. Thus, residual spatial autocorrelation was greatly reduced relative to OLS, although it was not fully eliminated. The negative sign indicates that the remaining spatial pattern is weakly negative rather than strongly positive.

Overall, the spatial lag model performed better than OLS across the main model comparison criteria. It produced a substantially lower AIC and BIC, a higher log-likelihood, and a significant likelihood-ratio test, while also reducing the magnitude of residual spatial autocorrelation. Although some heteroskedasticity, non-normality, and weak residual spatial dependence remained, the spatial lag specification represented a major improvement over the OLS model.

Spatial Error Regression Results

Table 3 - Spatial Error

```
Call:errorsarlm(formula = LNMEDHVAL ~ LNNBELPOV + PCTBACHMOR + PCTSINGLES +
  PCTVACANT, data = shp, listw = queenlist)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.926478	-0.115409	0.014889	0.133852	1.948663

Type: error

Coefficients: (asymptotic standard errors)

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	10.90643415	0.05346784	203.9812	< 0.000000000000000022
LNNBELPOV	-0.03453405	0.00708933	-4.8713	0.00000110883508
PCTBACHMOR	0.00981292	0.00072896	13.4615	< 0.000000000000000022
PCTSINGLES	0.00267792	0.00062083	4.3134	0.00001607386900
PCTVACANT	-0.00578307	0.00088670	-6.5220	0.0000000000006937

Lambda: 0.81492, LR test value: 677.61, p-value: < 0.0000000000000000222

Asymptotic standard error: 0.016373

z-value: 49.772, p-value: < 0.0000000000000000222

Wald statistic: 2477.2, p-value: < 0.0000000000000000222

Log likelihood: -372.6904 for error model

ML residual variance (sigma squared): 0.076551, (sigma: 0.27668)

Number of observations: 1720

Number of parameters estimated: 7

AIC: 759.38, (AIC for lm: 1435)

Likelihood ratio for spatial linear models

data:

Likelihood ratio = 677.61, df = 1, p-value < 0.000000000000000022

sample estimates:

Log likelihood of errreg	Log likelihood of reg
-372.6904	-711.4933

```

[1] 759.3807
[1] 797.5313
'log Lik.' -372.6904 (df=7)

Breusch-Pagan test

data:
BP = 23.213, df = 4, p-value = 0.0001148

studentized Breusch-Pagan test

data:
BP = 5.1627, df = 4, p-value = 0.271

Jarque Bera Test

data:  errreg$residuals
X-squared = 3507, df = 2, p-value < 0.000000000000000022

```

Table 3 presents the spatial error regression results. The key spatial parameter in this model is the error coefficient, λ , which captures spatial dependence in the unobserved portion of the model rather than in the dependent variable itself. In this case, $\lambda = 0.8149$ and is highly statistically significant ($p < 0.001$), indicating strong positive spatial autocorrelation in the error term. This suggests that unmeasured influences on median house values are spatially clustered across neighboring block groups. In other words, even after accounting for poverty, educational attainment, housing type, and vacancy, nearby areas still share omitted characteristics that affect housing values, and the spatial error model explicitly accounts for that structure.

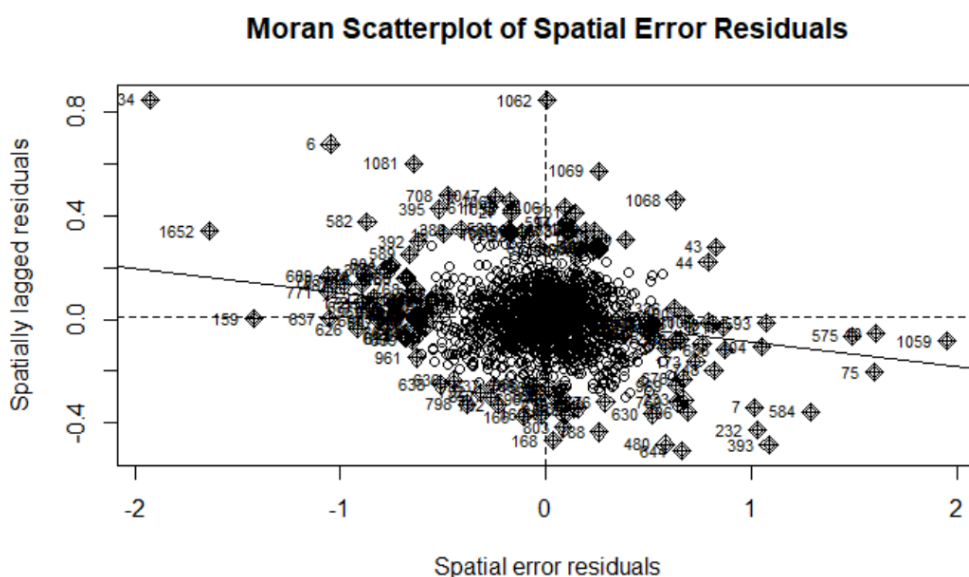
All four predictors remained statistically significant in the spatial error model. LNNBELPOV had a negative coefficient ($\beta = -0.0345$, $p < 0.001$), indicating that higher poverty levels were associated with lower median house values. PCTBACHMOR had a positive coefficient ($\beta = 0.0098$, $p < 0.001$), indicating that block groups with higher proportions of residents holding at least a bachelor's degree tended to have higher house values. PCTSINGLES was also positive and significant ($\beta = 0.0027$, $p < 0.001$), suggesting that a larger share of detached single-family housing was associated with higher median house values. PCTVACANT had a negative coefficient ($\beta = -0.0058$, $p < 0.001$), indicating that higher vacancy rates were associated with lower house values.

Compared with the OLS results, the signs of all coefficients remained the same, and all predictors were significant in both models. However, the coefficients in the spatial error model were smaller in absolute magnitude than in OLS. In the OLS model, the coefficients were -0.0789 for LNNBELPOV, 0.0209 for PCTBACHMOR, 0.0030 for PCTSINGLES, and -0.0192 for PCTVACANT, whereas in the spatial error model they were -0.0345, 0.0098, 0.0027, and -0.0058, respectively. This pattern suggests that part of the association estimated by OLS

reflected spatially clustered omitted influences, which are modeled more directly through the spatial error term in the spatial error specification.

Based on the Breusch-Pagan test, there is still evidence of heteroskedasticity in the spatial error residuals, although the result is weaker than in OLS and the spatial lag model. The standard Breusch-Pagan test remained significant ($BP = 23.213$, $p = 0.0001148$), indicating non-constant residual variance. However, the studentized Breusch-Pagan test was not statistically significant ($BP = 5.1627$, $p = 0.271$), which suggests that the evidence for heteroskedasticity is less consistent after the spatial error specification is applied. The Jarque-Bera test remained highly significant ($X^2 = 3507$, $p < 0.001$), indicating that the residuals were still not normally distributed.

Compared with the OLS model, the spatial error model showed a major improvement in fit. Its Akaike Information Criterion decreased from 1432.99 in the OLS model to 759.38 in the spatial error model, indicating substantially better model performance. The Schwarz Criterion (BIC) also decreased from 1460.24 in OLS to 797.53 in the spatial error model, again favoring the spatial error specification. The log-likelihood improved from -711.493 in the OLS model to -372.6904 in the spatial-error model. In addition, the likelihood ratio test comparing the spatial error model to OLS was highly significant ($LR = 677.61$, $p < 0.001$), confirming that the spatial error model provides a statistically meaningful improvement over the non-spatial OLS model.



The Moran scatterplot for the spatial error residuals shows much less spatial autocorrelation than the OLS residual plot. The residuals are more tightly concentrated around zero, and the fitted slope is far flatter, indicating that the spatial error model substantially reduced the strong positive spatial dependence present in the OLS residuals. This is supported by the Monte Carlo Moran's I

result for the spatial error residuals ($I = -0.0945$, $p = 0.002$). The remaining spatial pattern is weakly negative rather than strongly positive.

Overall, the spatial error model improved substantially on OLS. It reduced AIC, increased log likelihood, and significantly outperformed OLS in the likelihood ratio test. Although the remaining residual spatial autocorrelation was statistically significant, its magnitude was very small. This suggests that the model captured most of the spatial dependence, though some unobserved spatial error may still remain. Heteroskedasticity, non-normality, and weak residual spatial dependence were not fully eliminated.

Comparison of the Spatial Lag and Spatial Error Models

Since the spatial lag and spatial error models are not nested models, they were compared using information criteria such as the AIC and the BIC, where lower values indicate better model fit after accounting for model complexity. Based on these criteria, the spatial lag model performed better than the spatial error model. The spatial lag model produced an AIC of 525.48 and a BIC of 563.63, whereas the spatial error model produced a higher AIC of 759.38 and a higher BIC of 797.53. These results indicate that the spatial lag model provides a substantially better fit than the spatial error model and is therefore the preferred of the two spatial specifications.

3.4 – Geographically Weighted Regression Results

1. Present the global GWR results

Call:

```
gwr(formula = LNMEDHVAL ~ LNNBELPOV + PCTBACHMOR + PCTSINGLES +
     PCTVACANT, data = shps, gweight = gwr.Gauss, adapt = bw,
     hatmatrix = TRUE, se.fit = TRUE)
```

Kernel function: gwr.Gauss

Adaptive quantile: 0.008130619 (about 13 of 1720 data points)

Summary of GWR coefficient estimates at data points:

	Min.	1st Qu.	Median	3rd Qu.	Max.	Global
X.Intercept.	9.6727618	10.7143173	10.9542384	11.1742009	12.0831381	11.1138
LNNBELPOV	-0.2365244	-0.0733572	-0.0401186	-0.0126657	0.0948768	-0.0789
PCTBACHMOR	0.0010974	0.0101380	0.0149279	0.0202187	0.0347258	0.0209
PCTSINGLES	-0.0249706	-0.0075550	-0.0016626	0.0042280	0.0143340	0.0030
PCTVACANT	-0.0317407	-0.0142383	-0.0089599	-0.0035770	0.0167916	-0.0192

Number of data points: 1720

Effective number of parameters (residual: 2traceS - traceS'S): 360.5225

Effective degrees of freedom (residual: 2traceS - traceS'S): 1359.477

Sigma (residual: 2traceS - traceS'S): 0.2762201

Effective number of parameters (model: traceS): 257.9061

Effective degrees of freedom (model: traceS): 1462.094

Sigma (model: traceS): 0.2663506

Sigma (ML): 0.245571

AICc (GWR p. 61, eq 2.33; p. 96, eq. 4.21): 660.7924

AIC (GWR p. 96, eq. 4.22): 308.7123
 Residual sum of squares: 103.7248
 Quasi-global R²: 0.8479244

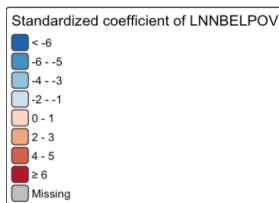
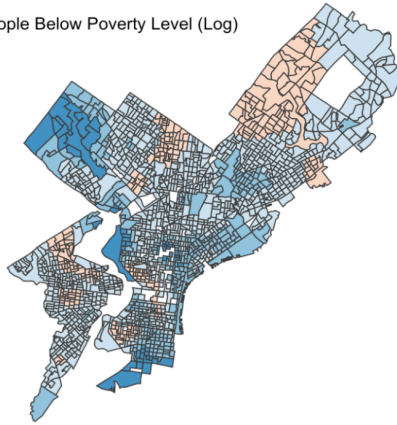
Since GWR regresses on each individual observation, the results presentation is not the same as OLS or Spatial Error where each predictor has a coefficient value. The GWR results instead present a range of coefficient values. The median coefficient for LNNBELPOV is -0.0401186, and 50% of the coefficient values fall between -0.0733572 and -0.0126657. The median coefficient for PCTBACHMOR is 0.0149279, and 50% of the values fall between 0.0101380 and 0.0202187. The median coefficient for PCTSINGLES is -0.0016626, and 50% of the coefficient values fall between -0.0075550 and 0.0042280. The median coefficient for PCTVACANT is -0.0089599, and 50% of the coefficient values fall between -0.0142383 and -0.0035770. Of all four of the predictors, only PCTBACHMOR's coefficient values are greater than 0 across all observations, whereas all the other 3 have maximum values greater than zero, but also have medians and minimum values less than 0. The median coefficient of each individual describes the typical relationship of that variable to the dependent variable, whereas the global coefficient simply fits a slope over all 1720 observations. The median and global coefficients for PCTBACHMOR are most similar, with a difference of around 0.00068, which points to a more normal distribution of coefficient values around the center. The global coefficient for LNNBELPOV is nearly two times the median coefficient (-0.0401186 median vs. -0.0789 global), suggesting a right-skewed distribution of coefficients. This suggests that the distribution of local coefficients is asymmetric, though the exact source of that asymmetry would require further investigation.

According to the R² values from the GWR and OLS regression outputs, the GWR model does a better job of explaining the variance of the dependent variable, LNMEDHVAL. The adaptive bandwidth GWR model has a quasi-global R² of 0.8479244, and the OLS regression model has an adjusted R² of 0.661513, meaning the GWR model accounts for 84.8% of the variance of log median home value, and the OLS regression model accounts for 66.2% of the variance of log median home value.

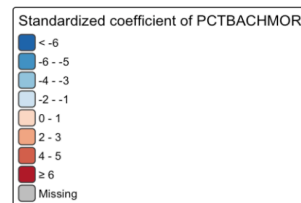
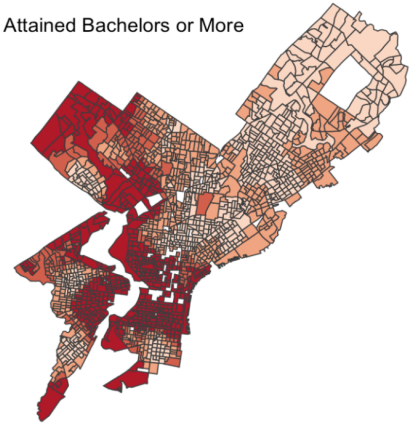
The Akaike Information Criterion (AIC) of the GWR model is 308.7123, and the AIC of the OLS regression is 1432.99. The GWR's drastically lower AIC suggests that despite its much greater complexity, GWR does a better job at fitting the data while also capturing more variation by spatial context across Philadelphia block groups. The AIC is a better measure of comparison for these models since the R² of OLS and the quasi-global R² of GWR are somewhat different statistics, whereas AIC is calculated with cross-model comparison in mind. Overall, the GWR model achieved a lower AIC, suggesting that despite being a much more complex model, it does a better job at capturing differences across spatial contexts as opposed to fitting one best fit line over all values of log median home value.

Local Regression Results

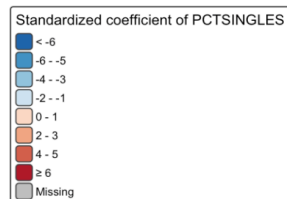
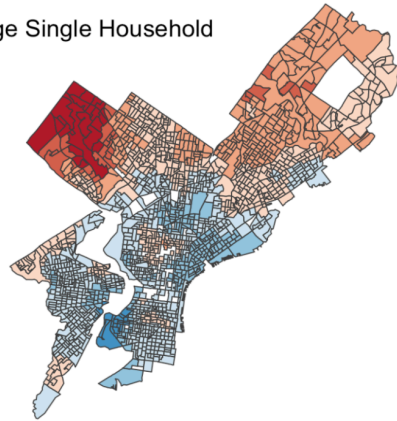
Number of People Below Poverty Level (Log)



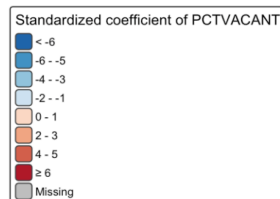
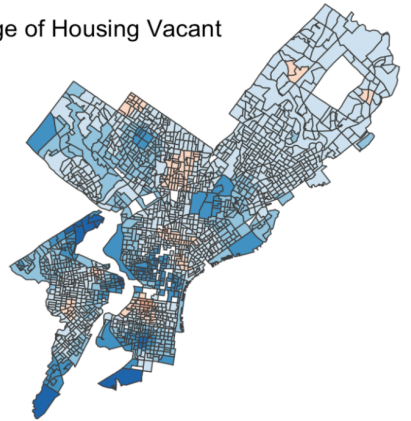
Percentage Attained Bachelors or More



Percentage Single Household



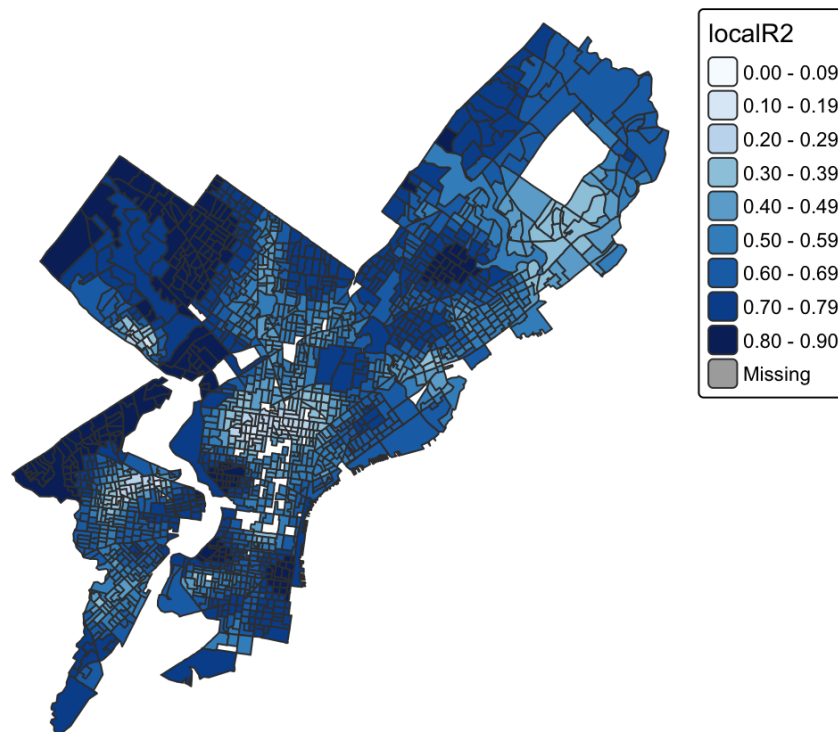
Percentage of Housing Vacant



Above are maps showing predictor coefficients divided by their standard errors. Greater absolute values (marked with darker blues and darker reds) mean coefficients have much greater absolute values than standard errors. When a coefficient is large relative to its standard error, there is stronger local evidence that the coefficient differs from zero. Lower values (between -2 and 2) suggest the local evidence for that coefficient is weaker. In the LNNBELPOV coev/SE map, there is a pattern of rather low coefficient: SE ratios across most of the city except for some parts of Northwest Philadelphia, near Fairmount Park, and in South Philadelphia, where

coefficient-to-standard-error (coef/SE) ratios are in the -4 to -3 range, which means LNNBELPOV shows stronger local evidence of association in those areas. In the PCTBACHMOR map, there is a strong concentration of high coef/SE in Center City, near West Philadelphia, and Northeast Philadelphia, where values are 6 or greater. In North and Northeast Philadelphia, as well as in pockets of South and Southwest Philadelphia, coef:SE ratios are closer to 0, suggesting that PCTBACHMOR is a less significant predictor in those areas. The PCTSINGLES map performs best as a predictor in Northwest Philadelphia where coef:SE ratios are 6 or greater. Otherwise, PCTSINGLES performs relatively well near Greys Ferry and the Devils Pocket and in Northeast Philadelphia, where coef:SE ratios are greater than 2 or less than -2. In the PCTVACANT map, most of the city, save a few spots in near Northeast, near North, and Southwest Philadelphia, has coef:SE ratios between -2 and 2, suggesting unreliable significance of the predictor for much of the city.

Across all four of the maps, North Philadelphia, more distant parts of West Philadelphia, and Northeast Philadelphia have lower coef:SE ratios, and therefore predictors struggle to predict as well in those areas. For all four predictors, Northwest Philadelphia boasts the greatest coef:SE ratios.



The choropleth map of local R-squares helps to understand where the predictor variables are best able to explain the variance of our dependent variable, LNLMEDHVAL. The greatest local R^2 values are highly concentrated in Northwest Philadelphia, South Philadelphia, University City

and northwest parts of West Philly, and select parts of Northeast and Center City. In these areas with the greatest R^2 values, most block groups have local R^2 between 0.7 and 0.9. Block groups with the worst local R^2 can be found north of Center City, and just northwest of University City, suggesting the model performs worst at explaining variation in the dependent variable (LNMEDHVAL).

4 – Discussion and Limitations

In this analysis, we examined spatial variation in median house values across Philadelphia block groups using LNMEDHVAL as the dependent variable and LNNBELPOV, PCTBACHMOR, PCTSINGLES, and PCTVACANT as predictors. We first reviewed the earlier OLS model and tested whether its assumptions were met, especially the assumption that residuals are spatially independent. LNMEDHVAL itself was highly spatially autocorrelated, and the OLS residuals remained clustered as well. While the OLS model explained a substantial portion of home value variation, the residuals still showed significant spatial structure.

Both spatial regression models improved on OLS. The spatial lag model performed especially well, with a much lower AIC and BIC than OLS, a much higher log likelihood, and substantially reduced residual spatial autocorrelation. The spatial error model also improved fit and captured most of the residual clustering left by OLS, but its AIC and BIC were higher than those of the spatial lag model, so among the two global spatial models, spatial lag was the stronger specification. However, the geographically weighted regression (GWR) produced the best overall performance in this report. In addition to its lower AIC and higher quasi-global R^2 , it also showed a smaller residual Moran scatterplot, which suggests that spatial autocorrelation had been reduced more than in the other models. Taken together, these results suggest that home values in Philadelphia are shaped not only by neighborhood socioeconomic conditions but also by spatial dependence and spatial non-stationarity. In other words, nearby home values matter, and the strength of predictor effects also varies across space.

At the same time, several limitations remained. Multiple assumptions were violated in the OLS model, such as spatial autocorrelation, heteroskedasticity, and non-normality. Although the spatial lag and spatial error models improved fit and greatly reduced residual spatial dependence, they did not fully resolve all of these issues. In the spatial lag model, heteroskedasticity, non-normality, and weak residual spatial autocorrelation remained. In the spatial error model, residual spatial autocorrelation was also reduced but not fully eliminated, and non-normality remained; heteroskedasticity was weaker than in OLS but still not entirely absent. GWR also has limitations, including sensitivity to local multicollinearity, small local sample sizes, and bandwidth choice. More broadly, all results depend on the Queen-1 weights matrix and the block-group unit of analysis, so different spatial weights or spatial units could lead to somewhat different findings.

It is also important to distinguish between spatially lagged residuals and spatial lag model residuals. Spatially lagged residuals are a diagnostic quantity created after fitting a non-spatial model such as OLS. They are produced by applying the spatial weights matrix to the residuals, so each observation is paired with a weighted average of neighboring residuals. If OLS residuals are correlated with their spatially lagged residuals, that indicates remaining spatial dependence and suggests that the OLS independence assumption has been violated. By contrast, spatial lag model residuals are simply the residuals that remain after estimating a spatial lag regression model. In that model, spatial dependence has already been incorporated directly into the regression equation through the spatially lagged dependent variable and the autoregressive parameter ρ . So, spatially lagged residuals are a diagnostic tool, while spatial lag residuals are the leftover errors from a fitted spatial lag model.

Finally, ArcGIS is somewhat problematic for GWR, which is why using R here is reasonable. Esri's ArcMap documentation says Local R^2 values from GWR should range from 0 to 1, yet ArcGIS users have reported negative local R^2 values in ArcGIS Pro output, which creates an inconsistency in interpretation. Because GWR is already sensitive to many factors, it is safer in this project to rely on R output that is easier to inspect and verify directly.